FRAM

FM24W256 - 256Kbit (32K x 8) I2C FRAM Module

v1.0

2025-09-26
Rev. A

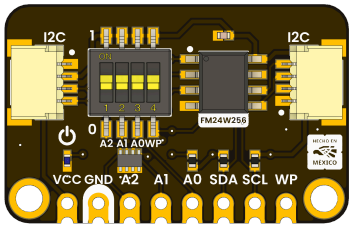
Professional electronic component

PRODUCT OVERVIEW

The FM24W256 module integrates a 256Kbit (32K x 8) I2C FRAM (Ferroelectric RAM) chip, providing a robust and efficient non-volatile memory solution for embedded applications. Designed for high-speed data access and low power operation, the module supports frequent and rapid read/write cycles with exceptional endurance. Unlike traditional EEPROM and other non-volatile memories, the FM24W256 ensures reliable data retention for up to 151 years, while minimizing system complexity and eliminating common reliability concerns. This makes it particularly suitable for data logging, configuration storage, and real-time data buffering in demanding embedded environments.

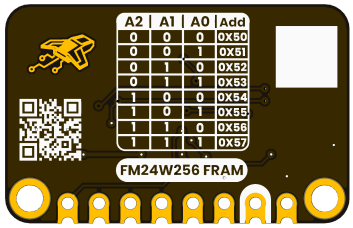
PRODUCT VIEWS

TOP VIEW



Component placement and connectors

BOTTOM VIEW



Underside components and connections

KEY TECHNICAL SPECIFICATIONS

🔌 CONNECTIVITY

Primary Interface:	**GPIO (Interrupt)**
Connector Type:	**JST 4-pin 1.0mm**
Logic Levels:	**VCC-referenced (2V – 5.5V tolerant)**

PIN CONFIGURATION

FUNCTION	NOTES
`Power Supply`	3.3V or 5V
`Ground`	Common ground for all components

KEY FEATURES

High Endurance

Supports up to 10^14 read/write cycles, far exceeding traditional EEPROM.

Fast Access Times

Typical write cycle time of 10 µs and read cycle time of 150 ns.

Low Power Consumption

Operates at 2.7

ADDITIONAL TECHNICAL INFORMATION

📦 OVERVIEW

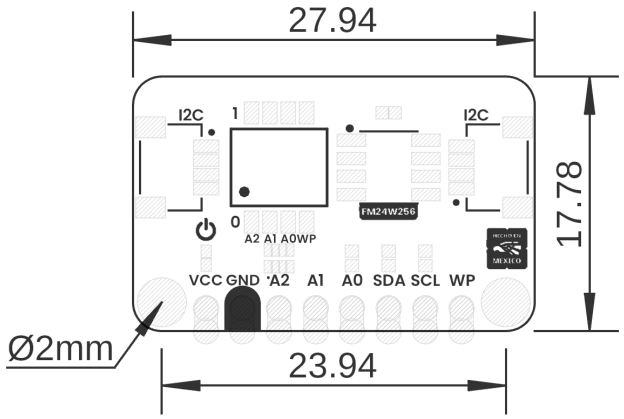
FEATURE	DESCRIPTION
Memory Type	FM24W256 I2C FRAM for dependable non-volatile storage
Interface	Standard I2C (2-wire) for straightforward integration
Operating Voltage	2.7V to 5.5V for flexible power compatibility
I2C Speed	Supports fast communication up to 400 kHz

TYPICAL APPLICATIONS

APPLICATION	DESCRIPTION
Data Logging	Ideal for storing sensor data and logs in real-time.
Configuration Storage	Retains system settings and configurations reliably.

HARDWARE DOCUMENTATION

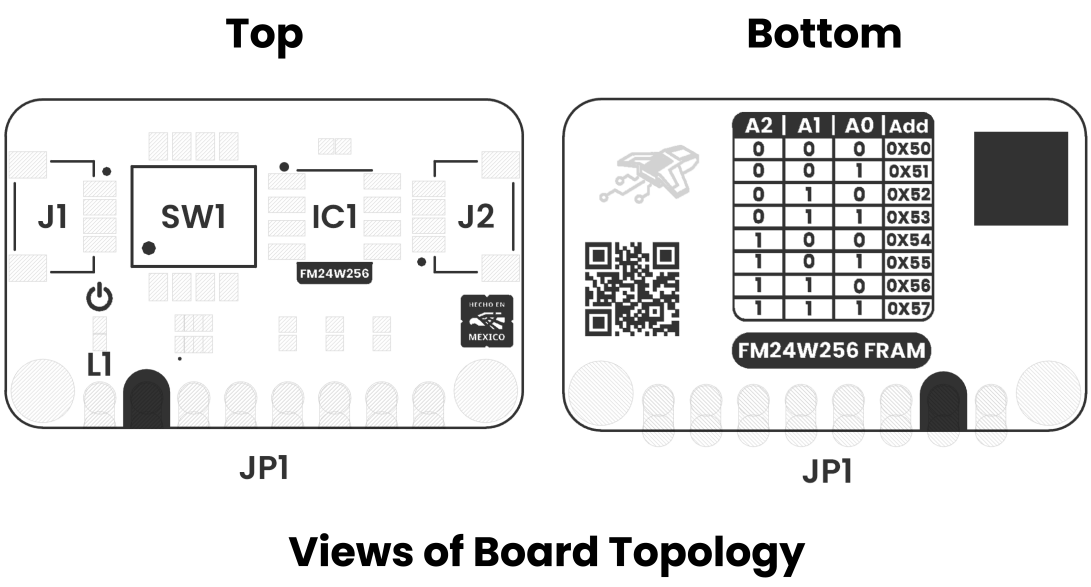
MECHANICAL DIMENSIONS



Board Dimensions in Millimeters

Physical dimensions and mounting specifications (measurements in millimeters)

SYSTEM TOPOLOGY



Views of Board Topology

Connection topology and system integration diagram

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COMPONENT REFERENCE

REF.	DESCRIPTION
IC1	FM24W256 FRAM
L1	Power On LED
JP1	2.54 mm Castellated Holes
SW1	Dip Switch for Configuration
J1	QWIIC Connector (JST 1 mm pitch) for I2C
J2	QWIIC Connector (JST 1 mm pitch) for I2C

CIRCUIT SCHEMATIC

Complete circuit schematic showing all component connections

View Complete Schematic PDF

PIN DESCRIPTION

Detailed pin assignment and electrical specifications

SIGNAL DESCRIPTION

FUNCTION	NOTES
Power Supply	3.3V or 5V
Ground	Common ground for all components

VOLTAGE LEVEL	FUNCTION
3.3 V – 5.5 V	Provides power to the on-board regulator and sensor core.
0 V	Common reference for power and signals.
1.8 V to VCC	Serial data line for I²C communications.
1.8 V to VCC	Serial clock line for I²C communications.

PIN CONFIGURATION LAYOUT

Physical connector layout and pin positioning

Pin Configuration Layout

Complete pin configuration diagram showing all connectors, pin assignments, and electrical connections for proper integration

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